



RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication
Engineering



Academic Year 2022-2023

Department of Electronics and Communication Engineering – Journal Publications

1. Vairaprakash S, Alagarsamy, M., Thangaraju, D., Mani, D.P. **Classification of COVID-19 Patient Based on Multilayer Perceptron Neural Networks Optimized with Garra Rufa Fish Optimization Using CT Scan Images** (2023) *International Journal of Pattern Recognition and Artificial Intelligence*, 37 (6), art. no. 2357003, . (SCI with IF:1.261)
2. Subha, T.D., Arunachalaperumal, C. **A hybrid CapSA-WHO method used for performance enhancement of tandem perovskite solar cell**(2023) *Expert Systems with Applications*, 230, art. no. 120554, . (SCI with IF: 8.665)
3. Savithri, M., Pradeepa, M., Prasad, D.R., Gangodkar, D., Rajalakshmi, R., Shafi, S., Sinchana, N.M., Prasanna Kumar, K.R., Selvam, N. **Enhancement of QoS in Internet of Things Wearable Devices Dependent on 5G Technology**(2023) *Wireless Communications and Mobile Computing*, 2023, art. no. 6709109, . (SCI with IF: 2.146)
4. Rajalakshmi, R., Sivakumar, P., Prathiba, T., Chatrapathy, K. **An energy efficient deep learning model for intrusion detection in smart healthcare with optimal feature selection mechanism** (2023) *Journal of Intelligent and Fuzzy Systems*, 44 (2), pp. 2753-2768. (SCI with IF: 1.737)
5. Periyannayagi, S., Nandini, V., Basarikodi, K., Sumathy, V. **IOT Assisted Biomedical Monitoring Sensors for Healthcare in Human** (2023) *Computer Systems Science and Engineering*, 45 (3), pp. 2853-2868. (SCI with IF: 4.397)
6. Nandagopal, C., Siva Kumar, P., Rajalakshmi, R., Anandamurugan, S. **Mobility Aware Zone-Based Routing in Vehicle Ad hoc Networks Using Hybrid Metaheuristic Algorithm** (2023) *Intelligent Automation and Soft Computing*, 36 (1), pp. 113-126. (SCI with IF: 3.401)
7. Thenmozhi, N., Perumal, B., Lakshmi, A. **Clustering-based image segmentation for optimal image fusion using CT and MRI images** (2023) *International Journal of Modeling, Simulation, and Scientific Computing*, art. no. 2441001, . (ESCI)
8. Sharma, A., Prasad, K.D.V., Chakrasali, S.V., Gowda V, D., Kumar, C., Chaturvedi, ., Pazhani, A.A.J. **Computer vision based healthcare system for identification of diabetes & its types using AI**(2023) *Measurement: Sensors*, 27, art. no. 100751, . (Scopus)
9. S. Vairaprakash, Jayakumar Karuppaiah, B. Buvaneswari, and Abdullah Alghamdi, **Removal of Salt and Pepper Noise Using Hybrid Adaptive Switching Median Filter with Ant Colony Optimization Technique in Nano Electronic Applications** (2023) *Journal of Nano electronics and Optoelectronics* <https://doi.org/10.1166/jno.2023.3365>. (UGC/Others)

10. B.Paulchamy, C.Natarajan, S.Vairaprakash, A.Purushothaman, **Implementation of Lane Marking Detection Algorithm for FPGA Using Soft Computing Technique** (2023), *Gradiva Review Journal*, DoI: 10.37897.GRJ.2022.V9I6.23.513143
11. R.Ramalakshmi , K.Ragavan , G.Gnana Priya , V.SrirengaNachiyar L.Krishna Kumari S.Tamil Selvi, **Performance Evaluation of Different MIMO Systems**, (2023) *Journal of Pharmaceutical Negative Results*, 10.47750/pnr.2023.14.S02.160 (UGC/Others)
12. Rajalakshmi R, Sivakumar P, Miroslav M and Muniyandy E **Adaptive Fuzzy Logic Deep-Learning Equalizer for Mitigating Linear and Nonlinear Distortions in Underwater Visible Light Communication Systems** (2023) *Sensors* 23(12), 5418 (SCI with Impact Factor: 3.847)
13. S.B. Mohan, B. Kannan, D. Ravikumar, C. Sivakumaran, **Hybrid Multi-kernel SVM Algorithm for microaneurysm recognition in Color Fundus Images** (2023), *International Journal of Medical Engineering and Informatics*, 10.1504/IJMEI.2023.10053229.(Scopus)
14. S. Durga Prasad, K.S.N. Murthy, B. Kannan, C. Sivakumaran, **Medical image registration and automatic hippocampus segmentation through convolutional neural network** (2023), *International Journal of Medical Engineering and Informatics*, DOI: 10.1504/IJMEI.2023.10055511. (Scopus)
15. KDV.Prasad, D.Pallanikumar, Dankan Gowda, Nazziya Hussain, A.Azhagu Jaisudhan Pazhani and Anil Kumar, **A Novel Method of Enhancing Security Solution and Energy Efficiency of IOT Protocols**(2023), *International Journal on Recent and Innovation Trends in Computing and Communication*, 11(4) pp (Scopus)
16. Bhuvaneswary.N, Deny.J, Lakshmi.A,**Hybrid optimized verification methodology using deep reinforcement neural network** (2023), *Journal of Intelligent &Fuzzy Systems*,DOI: 10.3233/JIFS-232132(SCI with IF: 1.737)
17. Ganesan, L., Umarani, C., Kaliappan, M., Vimal, S., Kadry, S., Nam, Y. **Texture Image Analysis for Larger Lattice Structure Using Orthogonal Polynomial Framework** (2022) *Information Technology and Control*, 51 (3), pp. 531-544. (SCI with IF: 0.813)
18. Lakshmi, A., Rajasekaran, M.P., Jeevitha, S., Selvendran, S. **An Adaptive MRI-PET Image Fusion Model Based on Deep Residual Learning and Self-Adaptive Total Variation** (2022) *Arabian Journal for Science and Engineering*, 47 (8), pp. 10025- 10042. (SCI with IF: 2.807)
19. B. Kannan, S. Karthigai Lakshmi, **Brain cancer analysis using deep learning architecture on MRI brain image**(2022), *International Journal of Medical Engineering and Informatics*, DOI: 10.1504/IJMEI.2022.10048344(scopus)
20. Azhagu Jaisudhan Pazhani, A., Samuel, T.S.A. **High-Speed and Area-Efficient Modified Binary Divider** (2022) *Circuits, Systems, and SignalProcessing*, 41 (6), pp. 3350-3371. (SCI with IF: 2.311)
21. Azhagu Jaisudhan Pazhani, A., Periyamayagi, S. **A novel haze removal computing architecture for remote sensing images using multi-scale Retinex technique**(2022) *Earth Science Informatics*, 15 (2), pp. 1147-1154. (SCI with IF: 2.705)

22. Ramkumar, M., Lakshmi, A., Pallikonda Rajasekaran, M., Manjunathan, A. **Multiscale Laplacian graph kernel features combined with tree deep convolutional neural network for the detection of ECG arrhythmia** (2022) *Biomedical Signal Processing and Control*, 76, art. no. 103639, . (SCI with IF: 5.076)
23. Jamuna Rani, M., Azhagu Jaisudhan Pazhani, A. **Computational efficient compression scheme for satellite images** (2022) *Earth Science Informatics*, 15 (3), pp. 1723- 1736. (SCI with IF: 2.705)
24. Singh, G., Appadurai, J.P., Perumal, V., Kavita, K., Ch Anil Kumar, T., Prasad, D., Azhagu Jaisudhan Pazhani, A., Umamaheswari, K. **Machine Learning-Based Modelling and Predictive Maintenance of Turning Operation under Cooling/Lubrication for Manufacturing Systems** (2022) *Advances in Materials Science and Engineering*, 2022, art. no. 9289320, . (SCI with IF: 2.098)
25. Gowda V, D., Sharma, A., L, R., Rahman, M., Yasmin, G., Sarma, P., Azhagu Jaisudhan Pazhani A., **A novel method of data compression using ROI for biomedical 2D images**(2022) *Measurement: Sensors*, 24, art. no. 100439, . (Scopus)
26. Periyannayagi, S., Gnana Priya, G., Chandrasekar, T., Sumathy, V., Raja, S.P. **Artificial intelligence and IoT-based biomedical sensors for intelligent cattle husbandry systems** (2022) *International Journal of Wavelets, Multiresolution and Information Processing*, 20 (6), art. no. 2250026, . (SCI with IF: 1.276)
27. Rajagopal, S., Ramakrishnan, T., Vairaprakash, S. **Anatomical Region Detection Scheme Using Deep Learning Model in Video Capsule Endoscope** (2022) *Intelligent Automation and Soft Computing*, 34 (3), pp. 1927-1941. (SCI with IF: 3.401)
28. Nallamilli P.G. Bhavani , Ganapathy Senthilkumar, Shahnazeer Chembalakkat Kunjumohamad , A.Azhagu Jaisudhan Pazhani , Ravi Kumar, Abolfazl Mehbodniya , Julian L. Webber, **Real-Time Inspection in Detection Magnetic Flux Leakage by Deep Learning Integrated with Concentrating Non-Destructive Principle and Electromagnetic Induction**(2022),*IEEE Instrumentation & Measurement Magazine*,25(7) pp 48-54. (SCI with IF: 1.548)
29. Azhagu Jaisudhan Pazhani A., Gunasekaran, P., Shanmuganathan, V., Lim, S., Madasamy, K., Manoharan, R., Verma, A. **Peer–Peer Communication Using Novel Slice Handover Algorithm for 5G Wireless Networks** (2022) *Journal of Sensor and Actuator Networks*, 11 (4), art. no. 82. (ESCI)
30. Naziya Hussain, Pullela SVVSR Kumar, Asan Mohideen Khansadurai, Dillip Narayan Sahu, A.Rameshbabu, Mirzanur Rahman, **IoT-Based Healthcare Application and Security Framework** (2022), *NeuroQuantology* 20(10), pp. 4428-4440 (UGC/Others)
31. Alagarsamy, M., Mathiyazhagan, V., Mani, D.P., Chitrakannu, M., Balaji Vignesh L.K., Suriyan, K. **Circular slot antenna for triband application** (2022) *International Journal of Reconfigurable and Embedded Systems*, 11 (3), pp. 226-232. (Scopus)

32. R.Deiva Nayagam, K.Aarthi, S.Mirra, **Colon Cancer Classification onHistopathological images using Deep Learning Techniques**(2022), *International Journal of Engineering Research & Technology*, 10(8) pp (UGC/Others)
33. Sivakumar G, Deiva Nayagm R, Balaji Vignesh L K, & Srirenga Nachiyar V,**Drug delivery system for chronic obstructive pulmonary diseases using IoT** (2022), *International Journal of Health Sciences*, 6(S8), 5331–5338.
34. Ravi Kumar, R. Prabha, B. Kannan, T.J. Nagalakshmi, Sheshang D. Degadwala, **Auditory model system to recognise Alzheimer's diseases: speech signal analysis**(2022), *International Journal of Medical Engineering and Informatics*, DOI: 10.1504/IJMEI.2022.10049259. (Scopus)
35. Gunasekaran, P., Azhagu Jaisudhan Pazhani A., Raj, T.A.B. **A Novel Method for Multiple Object Detection on Road Using Improved YOLOv2 Model** (2022) *Informatica (Slovenia)*, 46 (4), pp. 567-574. (ESCI)

